

are also available in 32- and 64-bit platforms. Select the type of file that suits your requirements.

Under Linux. Select the .tar.gz format file for Linux operating system (OS), 32-bit or 64-bit platform depending on the version of your OS. However, if you are not sure whether your OS is 32-bit or 64-bit, it is better to install the 32-bit version binary file. Installation in Linux is simple. Just unzip the file to target directory of your choice. Select the binary executable file Scilab by following the path in target directory scilab-5.x.x > bin > scilab. A script will be executed and then the main window will appear.

Under Windows. Select the suitable binary file for installation. Double click and run this file to open the installation wizard. You can either install the whole package or customise what all libraries need to be installed. For computers with Intel processors, installing Intel Math Kernel Library (MKL) increases the speed of computation.

Under Mac OS. The Mac version binary file for Scilab is in .dmg format. It uses Mac OS installer and follows the classic installation process familiar to Mac users.

Getting started

The main window, as shown in Fig. 1, is divided into four sub-sections: the Scilab console in the centre for making calculations, the files browser on the left, the variables browser on the top right and the work history on the bottom left.

TABLE I
System Requirements

GNU/Linux	➤ Debian, Redhat, Fedora, Suse and Ubuntu (32 and 64 bits) ➤ Pentium III class (or equivalent) with SSE instructions ➤ Minimum 1GB (preferably 2GB) RAM ➤ 550MB hard disk space Optional: ➤ Internet connection for ATOMS module installation ➤ C compiler (gcc/clang) ➤ Fortran compiler (gfortran)
Windows	➤ Windows XP/Windows Vista, Windows 7 and Windows 8 (32 and 64 bits) ➤ Pentium IV class (or equivalent) with SSE2 instructions ➤ Minimum 1GB (preferably 2GB) RAM ➤ 600MB hard disk space Optional: ➤ Internet connection for ATOMS module installation and Scilab installation with MKL ➤ C compiler (Visual Studio 2010 or Visual Express 2010)
Mac OS	➤ Leopard (10.5.x), Snow Leopard (10.6.x), Lion (10.7.x) and Mountain Lion (10.8.x) ➤ Mac Intel 64-bit processor ➤ Minimum 1GB (preferably 2GB) RAM ➤ 500MB hard disk space Optional: ➤ Internet connection for ATOMS modules installation ➤ C compiler (gcc/clang) ➤ Fortran compiler (gfortran)

In the console window, type any command after the prompt '-->' and press Enter key (in Linux and Windows) or Return key (in Mac OS) from the keyboard to display the result.

Basic numerical calculations

Basic arithmetic operations can be performed in Scilab with simple commands as given in the Table II. It is to be noted that the commands are case-sensitive. For example, only 'sqrt' is valid. 'Sqrt' and 'SQRT' show undefined variable and the calculation will not be performed.

Example. To find the cube, the product of numbers 3 and 5:

```
--> (3*5)^3
```

```
ans =
```

```
3375.
```

First type $(3*5)^3$ just after the prompt, as in the first line of the script given above. Press Enter/Return. The answer 3375 will be displayed in the console, as in the third line. You can navigate in the console using mouse or the ← → arrow keys of the keyboard.

If a semicolon (;) is added at the end of command line, the result would not be displayed in the console window, as given below:

```
--> (2+4)^2;
```

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